



your birth biomechanics handbook





Message from @birth-hood

Hi, I'm Leanne — mum of two (and two very excited cocker spaniels), Doula, Hypnobirthing teacher, Perinatal Yoga teacher, and, yes, I've been cheekily dubbed the "Queen of Biomechanics" by our local homebirth team!

I absolutely love helping parents understand how their bodies really work in pregnancy and birth. When you know how your pelvis, ligaments, and muscles move together, you can use positions, movements, and gentle exercises to make labour more efficient, reduce discomfort, and help your baby get into the best position for birth.

This is where birth ball work, perinatal yoga, and simple movement techniques come in. From rocking, leaning, and squatting to supported stretches =, use of rebozo and gentle strengthening, these tools help you work with your body, not against it. They can ease tension, open the pelvis, and encourage optimal fetal positioning, making birth feel smoother and more comfortable.

My goal is to empower you to move confidently, know which positions support you in labour, and feel strong, calm, and connected to your body — whether you're planning a home birth, birth centre, or hospital birth.

Leanne xx

Biomechanics is the study of biology and the mechanisms of movement.

Birth is not just about contractions, it's about movement. Your baby must navigate your pelvis, rotate, and flex to pass through the birth canal efficiently. This is where birth biomechanics comes in: the study of how your body and your baby move together during pregnancy and labour.

Knowing how your pelvis, spine, and ligaments work helps you:

- Position your body to give your baby more space to move
- Reduce labour pain and tension
- Support optimal fetal positioning
- Minimise interventions like assisted birth or caesarean

Simple tools, upright positions, gentle rocking, hip circles, hands and knees, or leaning forward can make a big difference. However understanding the mechanics of birth turns labour from a passive experience into an empowered, supported one, especially when adaptations are needed.

In childbirth, biomechanics and positional changes can help babies to rotate and turn to navigate the pelvis. Here's my guide to support you through pregnancy birth.

side lying release

Weekly from 34 weeks

Any stage of labour

How does it work?

This technique uses a 'static stretch' to temporarily enlarge and soften the pelvis. It can be done weekly to maintain balance and stability, help baby to rotate into an optimal position and encourage regular contractions. It can also ease back and labour discomfort.

How to do side-lying release:

- Begin by laying on a side of your choice. Your hips, shoulders, neck, and head should be inline.
- Your birth partner should stand in front of you for support.
- Move towards the edge of the bed until your bump extends over the edge.
- Flex your toes up towards your knees and allow your top leg to hang over the bed. Remain in this position for 3 contractions or 10 minutes.
- Repeat on the other side.

Avoid if you suffer from:

- Hypermobility
- Ehlers-Danlos Syndrome
- Severe pelvic or back pain



forward-leaning inversion

Daily from 32 weeks
Early or stalled labour

How does it work?

This technique increases room within the lower uterus for baby to move into an optimal position. It can be done in early labour, or when labour stalls. It can temporarily lengthen ligaments to aid alignment of the uterus with the pelvis. It may reduce back and hip pain and can help a breech or transverse baby to turn head down.

How to do forward-leaning inversion

- Kneel on the edge of a couch, bed, or chair.
- With the help of your partner, carefully lower yourself to your hands on the floor and then lower further to rest on your forearms.
- Keep your elbows out and hands close.
- Let your head hang freely. Your knees should be close to the edge, and your bottom up high.
- Take three breaths.
- With the support of your partner, come back up on to your hands, then up to a high kneeling position.
- Take two breaths here, then sit on your heels.



Avoid if you suffer from:

- High blood pressure
- Polyhydramnios (increased amniotic fluid)
- Any condition related to an increased risk of stroke

shaking the apple tree

Any stage of labour
Daily from 32 weeks

How does it work?

Jiggling activates stretch sensors in muscles, helping them to relax. Relaxed muscles create more space for babies to move. This can provide comfort in labour and encourage baby's head to descend into the pelvis in a good position. This can also help prevent a delay in labour.

How to do shake the apple tree:

- Adopt an all-fours position or lean over a counter.
- Using a rebozo, scarf or blanket, a birthing partner places the material, so it is hugging both buttocks.
- Holding the edges of the material at the hips, gently jiggle from left to right.
- Jiggle for up to 20 minutes, depending on comfort.
- Always check whether it is comfortable with the birthing person



Be cautious to
never shake the
hips vigorously

How does it work?

The sifting movement helps to relax abdominal muscles to encourage optimal fetal position. It can provide comfort in labour and encourage baby's head to descend into the pelvis.

How to do sifting

- Adopt an all-fours position.
- Using a rebozo, scarf or blanket, a birthing partner places the material, so it is hugging the tummy comfortably, top to bottom.
- Standing close to the birthing person's hips (to protect the partner's back), lift gently upwards and cycle the hands to create a rocking sensation.
- Cycle for up to 20 minutes, depending on comfort.
- Always check whether it is comfortable with the birthing person

Be cautious to
never shake the
hips vigorously



lunges

anytime in pregnancy
More useful in labour when
established or stalled

How does it work?

This technique aids rotation and descent of baby's head by opening the mid-pelvis. Lifting one leg can help create more room for baby to rotate into a good position. Can be used when labour has slowed and is a useful upright position to adopt in labour.

How to do a lunge:

- Whilst facing forwards, place one foot on a chair or stool to your side, toes pointing away from your body (90 degree angle)
- During a contraction, gently rock towards your bent knee beyond your foot.
- A birth partner can stand close or in front for support if needed.
- You can rest your foot on the floor between contractions if required.
- Try 5 contractions, then change to the other leg



Avoid if you suffer from

- You are struggling to stand on one leg
- In the presence of severe knee pain

counter pressure

From 34 weeks
Any stage of labour

How does it work?

This technique involved a pelvic/hip squeeze/pressure which can help ease the discomfort of contractions, especially for a birthing person experiencing back labour or intense pressure in their pelvis.

How to do counter pressure:

Sacral pressure:

- Feel for the triangular bone on the back of the pelvis.
- Placing your entire palm flat onto the bone (there may be a tendency to put more weight into the heel of the palm), you will press directly down into the sacrum.

Hip squeeze:

- Find the two bony points at the front of the pelvis.
- Place the heels of your palm on the side of the hip/pelvis (finger tips pointing towards each other, and squeeze during the duration of the contraction.
- The goal is to press the hips in and towards each other to help relieve pain during a contraction.



Considerations:

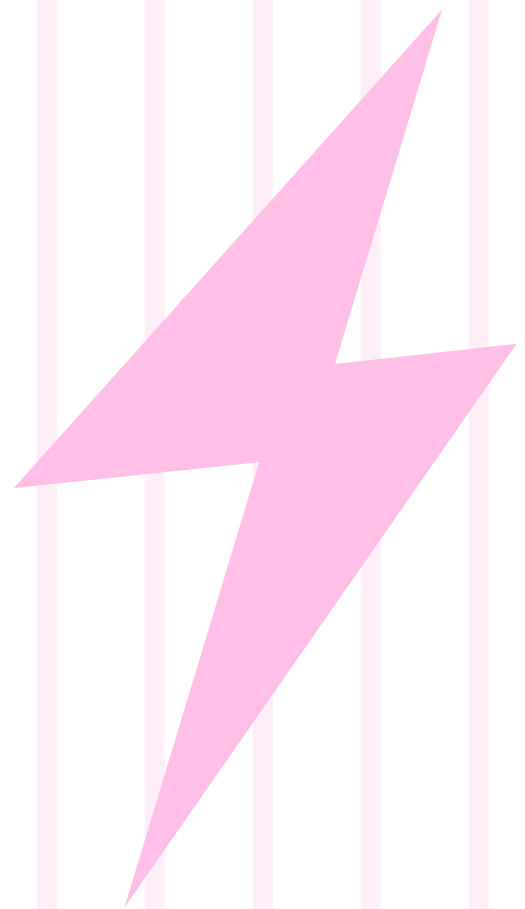
- Applying counter pressure can relieve pain from a contraction, but if the pain is from baby's position being slightly off or less optimal, we also want to work to correct the position.

alternatives

There are many other things which can help you to stay relaxed and comfortable in pregnancy and labour.

These include:

- Massage
- TENS (electrical nerve stimulation), TENS machines can be hired or bought privately
- Changing position
- Warm or cool compress
- Peanut ball / birth ball
- Aromatherapy
- Distraction
- Favourite foods
- Hypnobirthing
- Bath or shower
- Low lighting
- Music
- Sitting on the loo/birth stool
- Accupressure
- Vocalising
- Breathing





hip squeeze

Lean forward while your partner gently squeezes your hips inward and slightly upward during contractions or moments of tension.



lunges

Step one foot forward into a supported lunge and slowly sink into the movement, breathing deeply and switching sides after 5 surges.



shaking the apple tree

Lean forward or come onto all fours while your partner uses a scarf around your tummy to gently jiggle you from side to side taking the weight of the bump.



sifting

Lean forward or come onto all fours while your partner uses a scarf around your hips to gently jiggle your pelvis from side to side.



birth ball tilts

Sit on a birth ball and slowly tilt your pelvis forwards and backwards, creating gentle rocking movements through your hips.



forward leaning inversion

Lean your upper body forward over a sofa with your hips lifted and hold for 3-5 breaths before carefully returning upright kneeling.



sacral counter pressure

Lean forward while your partner applies firm, steady pressure to your lower back and sacrum to ease discomfort and encourage pelvic space.



side lying release

Lie on your side with your top leg hanging over the edge of the bed while your partner supports you, allowing your pelvis to soften and release for 3 contractions on each side.



useful resources/ sources

www.spinningbabies.com



www.optimalbirth.co.uk



leanne@birth-hood.co.uk

www.birth-hood.co.uk

07814504865